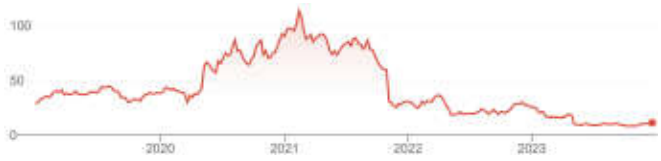

FIG. 3. Historic share prices in USD of Chegg (NYSE CHGG) as a proxy for the popularity of online problem-solving sites. Prices greatly increased when courses went online as a result of the pandemic, but fell again dramatically coinciding with the emergence of Large Language Models.

  - You used Internet resources like help sites or forums to solve them (*OnlInt*).

- Your Opinion: Please tell us what you think about using AI in online classes; what should ideally be done; what should not be done?

The survey was administered online during the last week of the semester, but results were not viewed or analyzed until the grades for the course had been turned in. A nominal participation credit was given for submitting the survey, regardless of whether or not the students agreed to be part of the study. The students were aware of this protocol as part of the informed consent, and data was only analyzed for students who agreed to participate. The study was approved under MSU-IRB-STUDY00009987.

B. Considered variables

We compiled a range of variables that capture various aspects of student performance and behavior, shown in Table I. Key performance metrics include *Hwk* (homework score), *OnlExams* (score from online exams), *CamExams* (score from on-campus exams with supervision), and *Final* (score from the final exam, also conducted on-campus with supervision). These scores are presented as percentages, reflecting the students' achievement in each respective assessment. Additionally, *Sem5* represents the scores for five problems initially available online, *Final5* for the same problems when included in the final exam, and *Diff5* indicating the score difference between these two settings. Thus, *Diff5* can be used as a proxy for retention of concepts between the semester and the final exam. Finally, *DiffExams* quantifies the score difference between online and on-campus quizzes, offering insight into performance variations across different assessment environments.

The dataset also encompasses variables related to the use of digital resources and student interactions. *HwkAI* and *OnlAI* denote the self-reported percentage of problems for which artificial intelligence (AI) tools were used during homework and online quizzes, respectively. Similarly, *HwkInt* and *OnlInt* represent the usage of other internet resources in these settings. Finally, *HwkPeer* and *HwkFac* quantify the self-reported extent of peer discussions and interactions with faculty during homework.

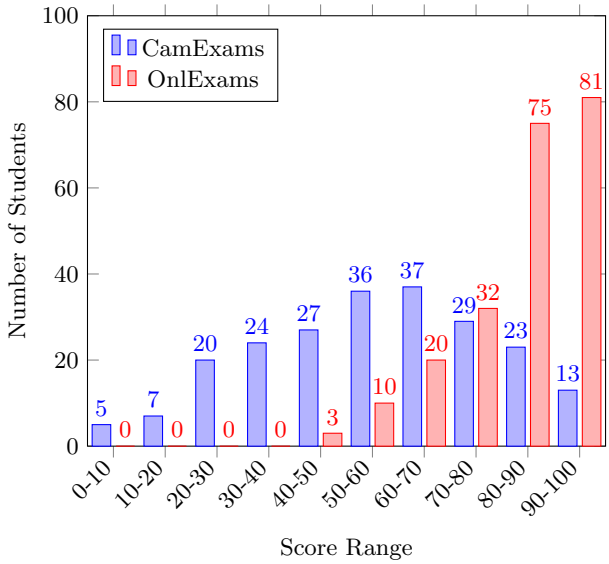

FIG. 4. Comparison between score distributions for the nine low-stakes exams that were conducted online (red) and the two low-stakes exams that were conducted on-campus under supervision (blue).

C. Statistical methods

Data were downloaded from the course management system, and survey results were merged using Python scripts. Calculations for this project were carried out using ChatGPT-4 Advanced Data Analysis [15] and R [21] (in particular qgraph [22] and CTT [23]).

IV. RESULTS

A. Response rate

The first and second semester courses were completed by 156 and 183 students, respectively. Of these, 90 and 131 students agreed to participate in the study, bringing the total to 221 participants.

B. Online versus on-campus exams

Figure 4 shows the score distributions for the nine exams that were conducted online and the two exams that were conducted on-campus under supervision. In a t-test, these distributions are significantly different ($p \approx 4.7 \cdot 10^{-47}$).

The conditions under which these exams were conducted led to vastly different outcomes, and an immediate assumption would be that this is related to the use

TABLE I. Summary of variables in the dataset

Variable	Description	Min	Max	Mean	Std. Deviation
CamExams	On-Campus Exam Score (%)	0.00	100.00	53.80	22.73
Diff5	Difference in Online/Final Duplicate Problem Scores	-100.00	40.00	-39.00	33.80
DiffExams	Difference in Online/On-Campus Exam Scores	-81.55	23.57	-29.30	20.04
Final	Final Exam Score (%)	0.00	100.00	49.19	27.64
Final5	Final Exam Duplicate Problem Score (%)	0.00	100.00	51.22	33.37
Hwk	Homework Score (%)	25.00	100.00	86.60	15.68
HwkAI	AI Usage During Homework (%)	0.00	100.00	17.08	23.94
HwkFac	Discussions with Faculty During Homework (%)	0.00	100.00	16.32	24.29
HwkInt	Internet Usage During Homework (%)	0.00	100.00	50.00	30.18
HwkPeer	Peer Discussions During Homework (%)	0.00	100.00	32.79	31.44
OnlAI	AI Usage During Online Exams (%)	0.00	88.00	8.07	17.54
OnlExams	Online Exam Score (%)	43.69	98.06	83.10	11.77
OnlInt	Internet Usage During Online Exams (%)	0.00	100.00	23.38	29.39
Sem5	Online Problem Score (%)	20.00	100.00	90.23	17.33

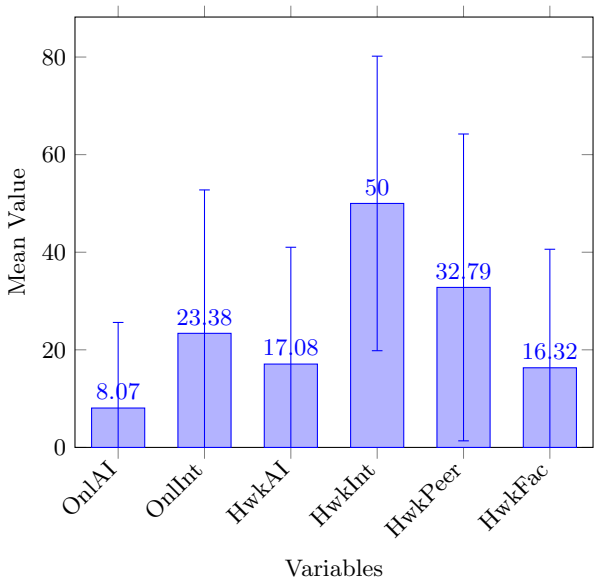

FIG. 5. Self-reported usage of resources while working on homework and online exams (see Table I for the explanation of the variables).

of external resources during unsupervised assessments. As an example, for nine of the ten questions on the last online exam, solutions could be found on Chegg within about one minute each.

C. Usage of resources during unsupervised assessments

Figure 5 illustrates the average self-reported usage of AI and other internet resources, as well as self-reported consultation with peers and course personnel.

Overall, students report less usage of resources during exams than during homework, but not significantly. The only one-sigma significant differences are between usage of AI and talking to faculty on the one hand, and using other internet resources while working on homework; students have not yet adopted AI and stick with "traditional" problem-solving sites. On the average, students use other internet resources for half of the homework problems. The distributions of these types of resources usages, however, are very different, see Fig. 6, which suggests that there a different classes of resource usage.

Using k-means clustering and elbow method, we identified four different classes as indicated in Table II. Cluster 1, the smallest group, is comprised of students who appear to prefer human interaction to any online resources, and these students mostly adhere to rules for the online exams. Students in Cluster 2, the largest cluster, state that they make little use of resources overall, and that they most closely adhere to rules for the online exams. Students in Cluster 3 make heavy use of internet resources other than AI in both homework and exams, thus not following rules. Finally, students in Cluster 4 use all available resources and disregard exam rules.

D. Correlations of attributes

Figure 7 shows a Fruchterman-Reingold [22, 24] representation of the correlation matrix between the variables. Indicated in light blue are the online, unsupervised assessments, in green the on-campus, supervised assessments, and in gray the differences in scores between selected subsets of assessments. The percentages of AI-usage are indicated in beige, usage of internet resources in yellow, and discussions with humans in orange. Green edges denote positive correlations, red edges negative correlations, and the thickness their absolute strength. Due to the force-directed nature of Fruchterman-Reingold graphs, closely correlated vertices tend to cluster, while unrelated vertices tend to be further apart from each other.

It is apparent that the scores achieved online and those

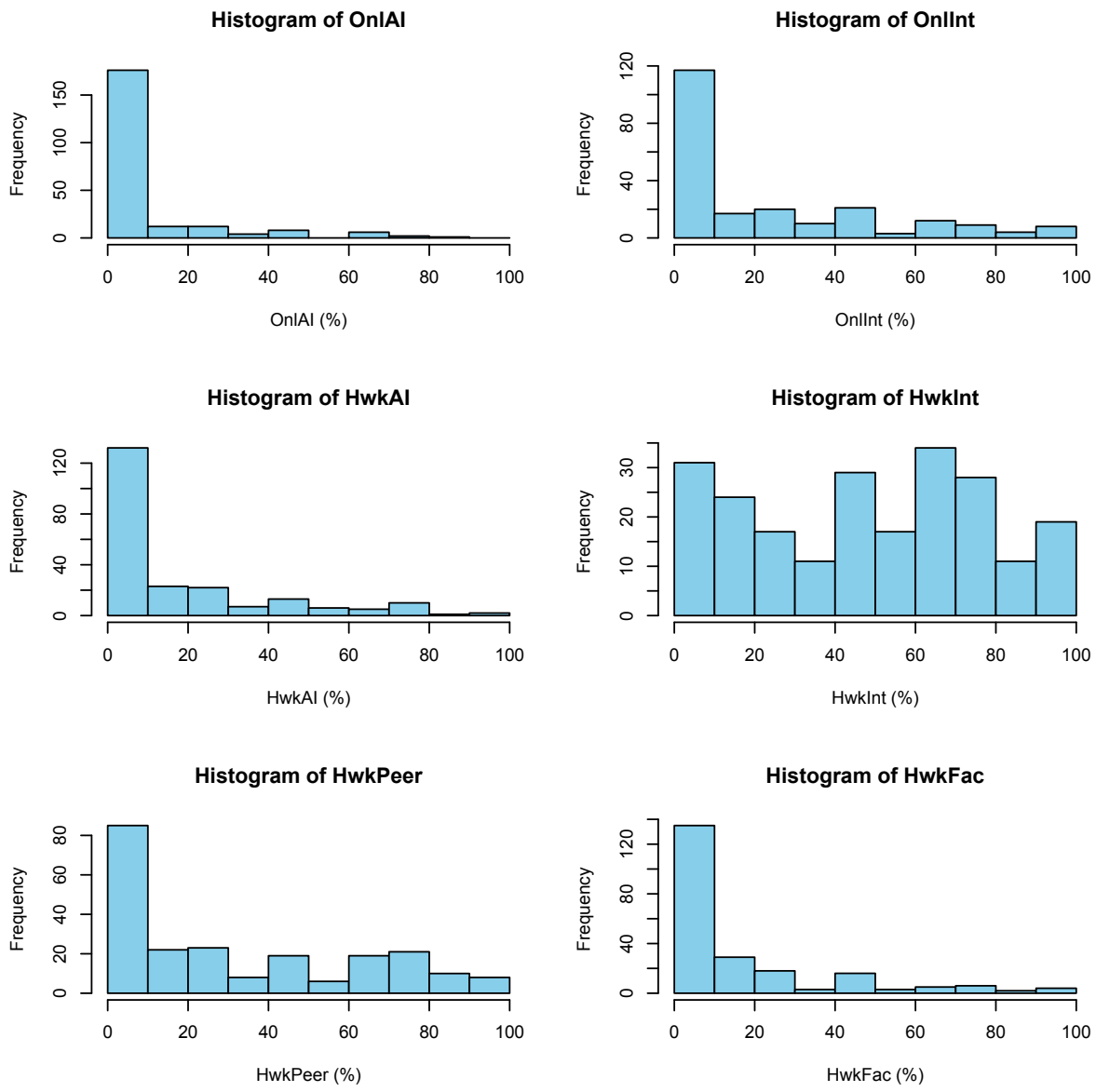

FIG. 6. Distribution of the survey responses.

TABLE II. Number of members and mean values of variables in identified clusters of resource usage.

Cluster	# members	*HwkAI*	*HwkInt*	*HwkPeer*	*HwkFac*	*OnlAI*	*OnlInt*	Interpretation
1	27	9.6	39.9	53.1	66.4	3.6	14.3	Using mostly peer discussions and helprooms
2	96	8.3	29.3	20.6	5.7	1.2	3.1	Using resources on homework but not exams
3	56	5.1	80.1	35.8	7.4	1.9	51.3	Heavily using internet on homework and exams
4	42	58.0	63.7	43.6	20.3	34.8	38.2	Using all resources including AI everywhere

achieved under supervision each cluster together, but both are disconnected from the survey answers. Taking *Final* as a proxy for learning success in the course, after discarding the derived variable *DiffExams* and all variables related to the five duplicate problems, in a linear regression of all remaining assessment and survey variables, only *CamExam* emerges as a statistically significant predictor of the final exam score ($p \approx 7.3 \cdot 10^{-16}$); *OnlExams* comes close to statistical significance with $p = 0.05$. Table III shows the outcome of this linear regression, which has a week correlation of $R^2 = 0.36$.

Also within the usage classes (Table II), there are very few significant correlations between resource usage and assessment performance. Figure 8 shows the statistically

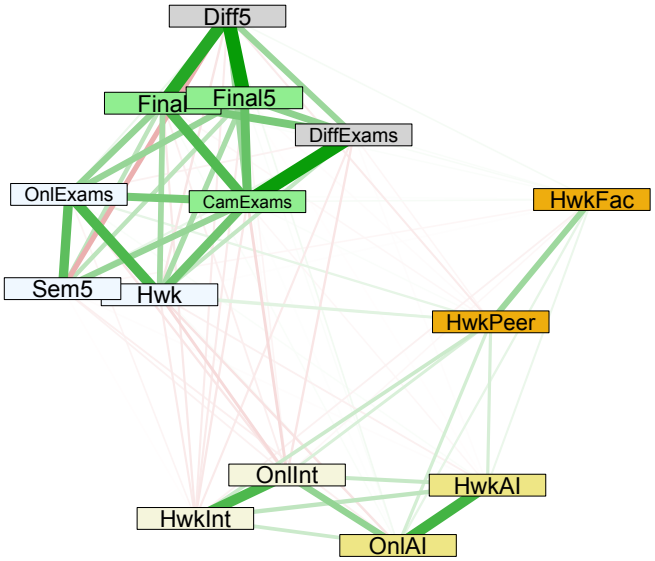

FIG. 7. Fruchterman-Reingold [22, 24] representation of all correlations between the variables in Table I for all students.

TABLE III. Linear regression for *Final*.

	Estimate	Std. Dev.	t	p
Hwk	−0.08	0.13	−0.6	0.55
OnlExams	0.33	0.17	1.96	0.05
CamExams	0.69	0.08	8.74	$7.31 \cdot 10^{-16}$
HwkAI	−0.06	0.09	−0.68	0.50
HwkInt	−0.05	0.07	−0.70	0.49
HwkPeer	−0.04	0.05	−0.78	0.44
HwkFac	0.01	0.07	0.20	0.84
OnlAI	0.06	0.12	0.46	0.65
OnlInt	0.05	0.07	0.74	0.46

significant correlations between the variables ($p < 0.05$). For the heavy internet users (Cluster 3), the usage of internet resources other than AI during online exams (*OnlInt*) is significantly negatively correlated with the scores on the exams (*OnlExams*) ($r = -0.28$; $p = 0.04$), which may indicate that relying on the internet, the students were not able to quickly enough find what they needed to correctly solve the problems, including replacement of the numbers by their values. For the users who made use of all resources everywhere (Cluster 4), the use of AI during online exams (*OnlAI*) is significantly positively related to *Diff5* ($r = 0.31$; $p = 0.04$); this means that AI-usage during online exams is positively correlated with doing better on the final exam instance of duplicate problems than on their first occurrence in the course.

Notably, within all usage classes, significant correlations between the supervised final exam and the unsupervised assessment components of the course remain. This means that in spite of even the heaviest use of external resources, if not as a significant predictor, unsupervised assessment still retains formative relevance.

TABLE IV. Average problem solving mean and average point biserial correlations for homework and online exams by clusters.

	Homework		Online Exams	
Cluster	Mean	pBis	Mean	pBis
1	0.87	0.46	0.82	0.33
2	0.86	0.45	0.82	0.30
3	0.84	0.40	0.81	0.28
4	0.85	0.49	0.83	0.26

Using *Final* as a proxy for learning success, we also investigated if higher or lower performing quartiles of the students may have benefitted or been harmed by the use of online resources, but we found no difference in that regard between these populations.

E. Item analysis

The use of external resources is detrimental to the validity of assessments. Table IV shows the average item parameters for assessment items on homework and unsupervised online exams. The mean reflects the average percentage of correctly solved items, and the point biserial ("pBis") the discrimination of these items (ranging from −1 to 1 where negative values usually denote invalid assessment items), that is, how well the item distinguishes between students who generally have a good grasp of the concepts and those who do not.

The values only insignificantly differ between different classes of online resource usage patterns. Overall, they are consistent with the low predictive power of these unsupervised assessments, but they still indicate that items can provide feedback to students and instructors.

F. Student comments about AI

Based on the replies to the open-ended question on the survey, many students recognize AI as a valuable tool for assisting in learning, particularly for understanding complex topics and guiding problem-solving. They particularly value its ability to quickly provide information without having to flip through textbook materials or scroll through video. They appreciate AI's ability to provide alternate explanations and solutions, which can be especially helpful when traditional teaching methods fall short. However, there's a consensus that AI should not replace genuine learning and effort. Students suggest that AI's role should be that of an assistant rather than a solution provider, and its usage should be context-dependent. For instance, in major-related courses, students advocate for minimal AI use to ensure a solid understanding of essential concepts. Conversely, in subjects that are less critical for them, they see AI as a more acceptable aid.

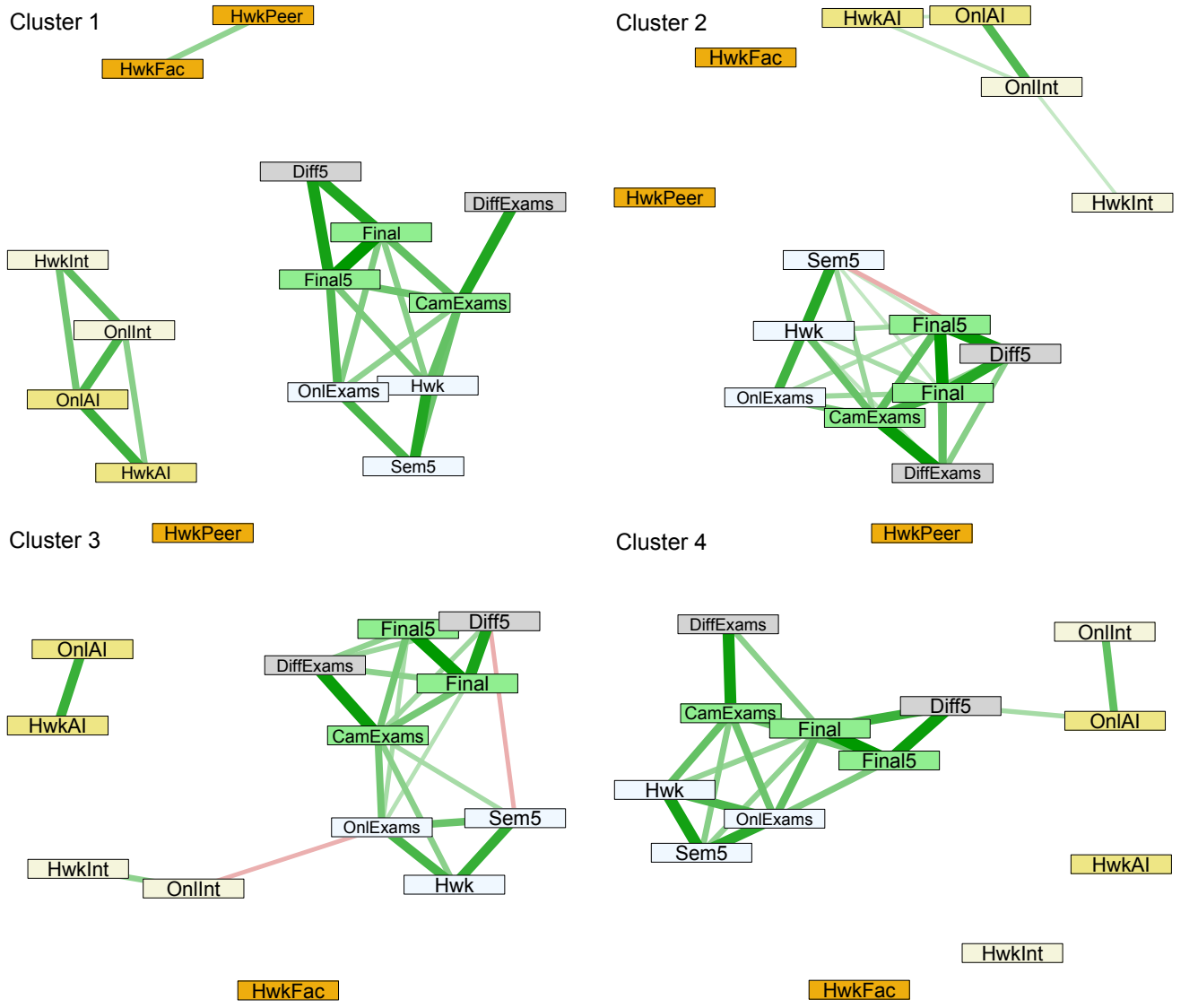

FIG. 8. Fruchterman-Reingold [22, 24] representation of the statistically significant correlations ($p < 0.05$) between the variables in Table I for the clusters in Table II. Note the rotation and handedness of these representations are random.

Concerns about academic integrity and the potential for AI to promote laziness and dependency are prominent. Students worry that reliance on AI for problem-solving or essay writing could lead to a superficial understanding of course material and hinder the development of critical thinking skills. They propose a balanced approach, where AI is used judiciously to enhance learning without becoming a crutch. This balance involves using AI for initial guidance or concept clarification while avoiding its use for directly solving assignments or exams. Some students commented on having some of the exams during the semester being in-person as beneficial. Ironically, but of course also very temptingly, based on style considerations, it appears that several students filled out the free-response question using ChatGPT.

Finally, the practicality of regulating AI use in online education is a significant concern. Some students acknowledge that while AI tools like ChatGPT may not be sophisticated enough currently to solve complex academic problems accurately, they could still be misused. Several students find ChatGPT is not yet trustworthy, but expect this to improve in the future. There's an acknowledgment that AI is a part of the evolving educational landscape, and rather than outright banning it, educators should find ways to integrate it responsibly into the curriculum. This integration could involve designing assessments that still require a deep understanding of the material, even with AI assistance, and teaching students how to use AI ethically and effectively as part of their learning toolkit.